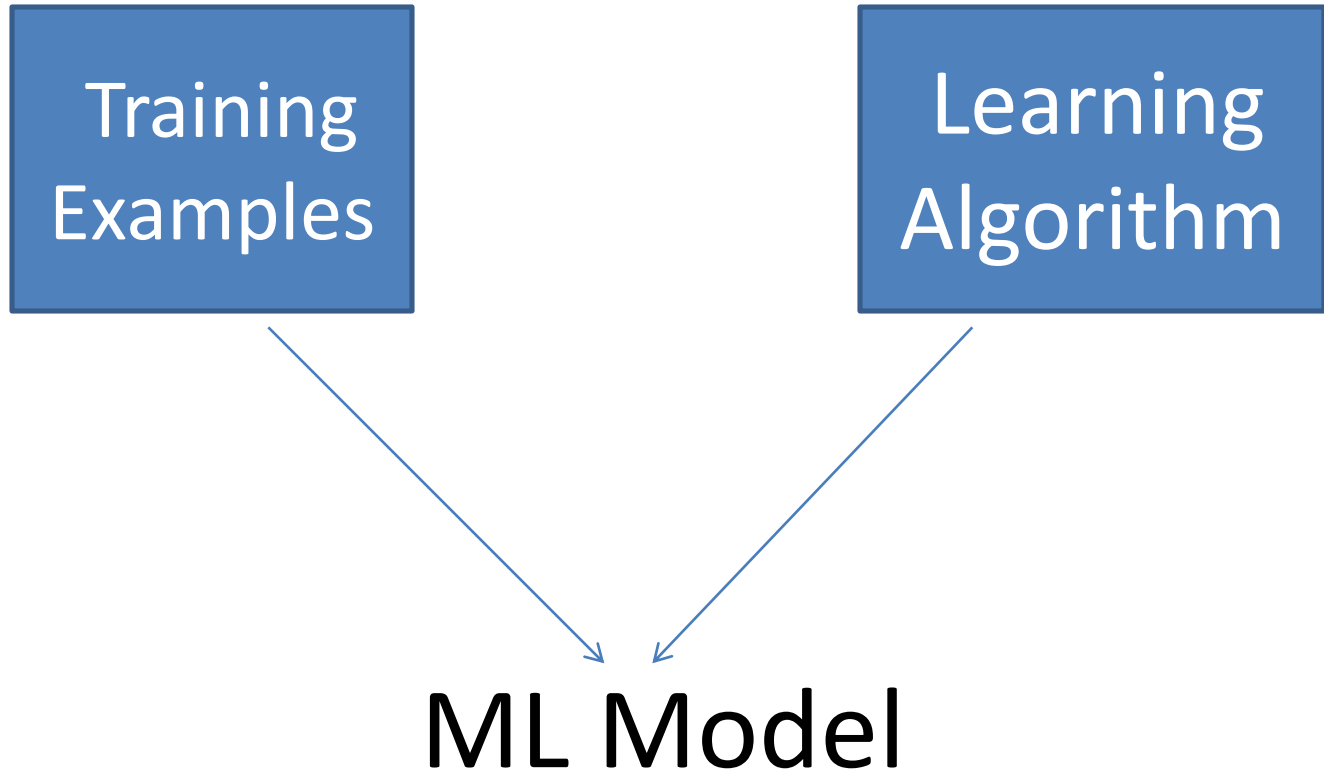
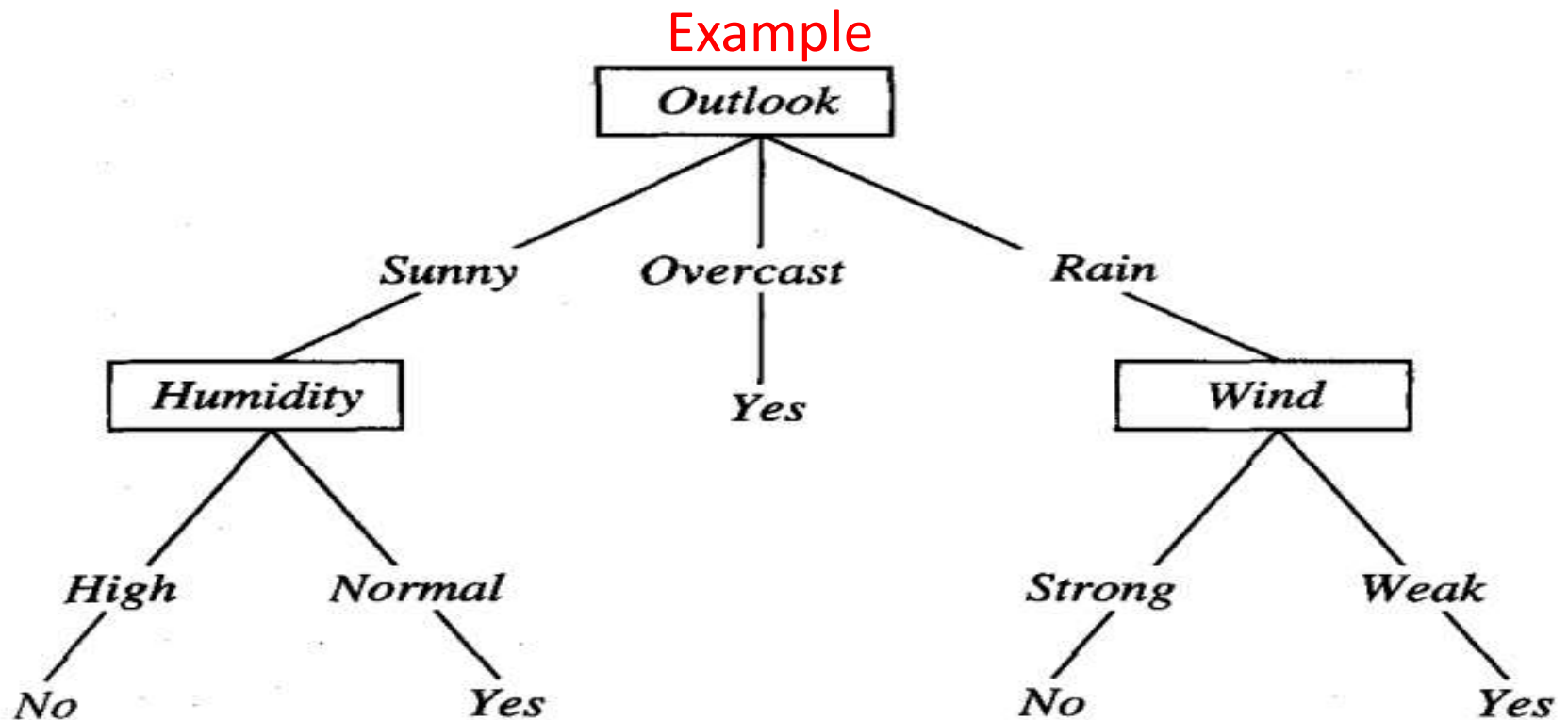


Decision Tree



Decision trees classify instances by sorting them down the tree from the root to some leaf node, which provides the classification of the instance. Each node in the tree specifies a test of some attribute of the instance, and each branch descending from that node corresponds to one of the possible values for this attribute.



Day	<i>Outlook</i>	<i>Temperature</i>	<i>Humidity</i>	<i>Wind</i>	<i>PlayTennis</i>
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$Entropy(S) \equiv \sum_{i=1}^c -p_i \log_2 p_i$$

$$Entropy(S) \equiv -p_{\oplus} \log_2 p_{\oplus} - p_{\ominus} \log_2 p_{\ominus}$$

$$Gain(S, A) \equiv Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$\begin{aligned} \text{Entropy}([9+, 5-]) &= -(9/14) \log_2(9/14) - (5/14) \log_2(5/14) \\ &= 0.940 \end{aligned}$$

$$\text{Values}(\text{Wind}) = \text{Weak}, \text{Strong}$$

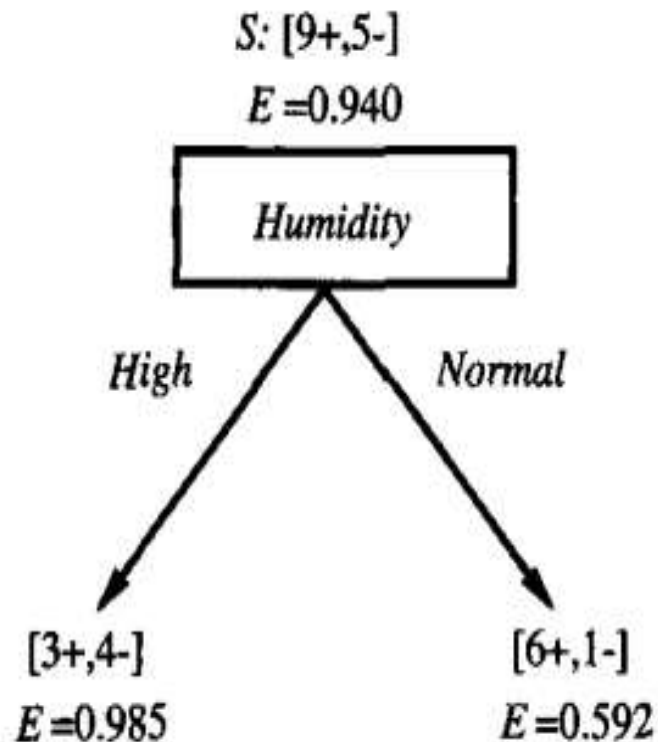
$$S = [9+, 5-]$$

$$S_{\text{Weak}} \leftarrow [6+, 2-]$$

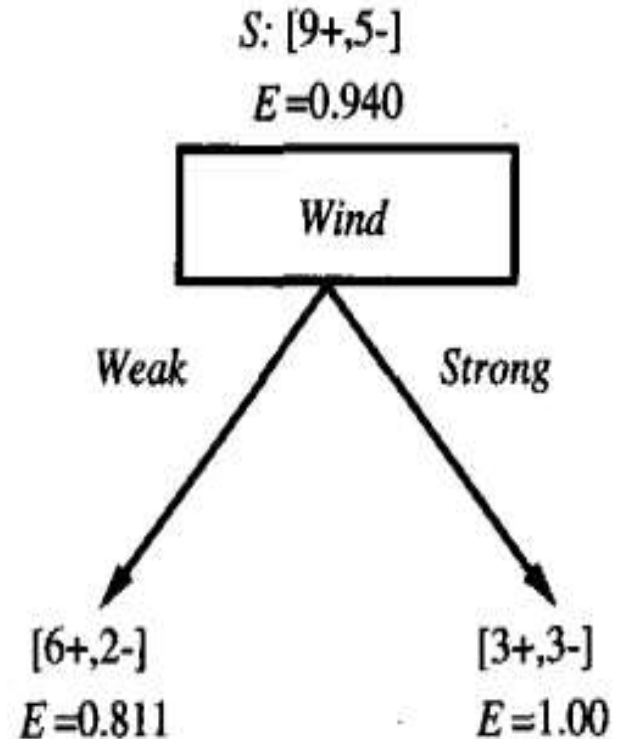
$$S_{\text{Strong}} \leftarrow [3+, 3-]$$

$$\begin{aligned} \text{Gain}(S, \text{Wind}) &= \text{Entropy}(S) - \sum_{v \in \{\text{Weak}, \text{Strong}\}} \frac{|S_v|}{|S|} \text{Entropy}(S_v) \\ &= \text{Entropy}(S) - (8/14) \text{Entropy}(S_{\text{Weak}}) \\ &\quad - (6/14) \text{Entropy}(S_{\text{Strong}}) \\ &= 0.940 - (8/14)0.811 - (6/14)1.00 \\ &= 0.048 \end{aligned}$$

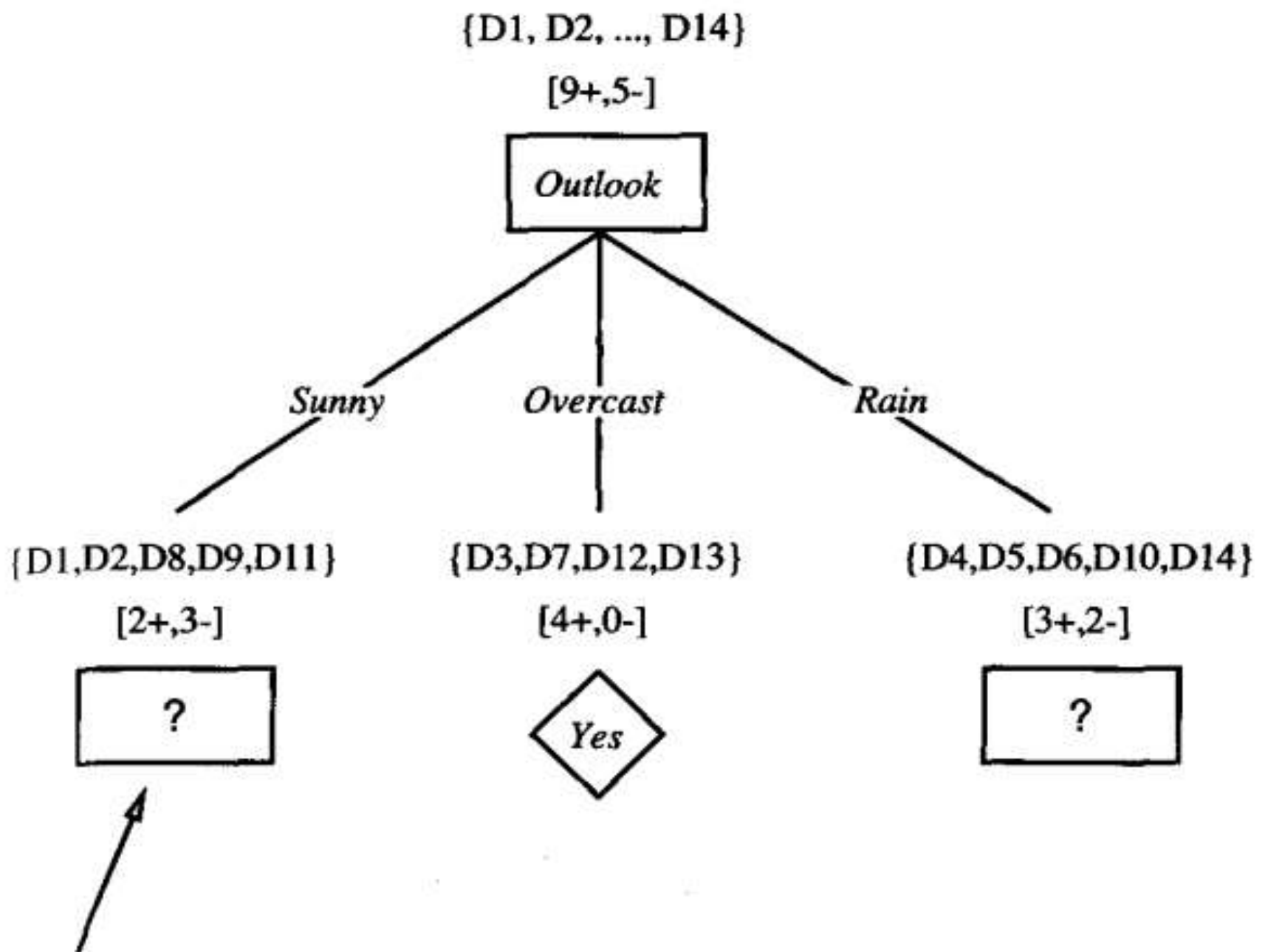
Which attribute is the best classifier?



$$\begin{aligned} \text{Gain}(S, \text{Humidity}) &= .940 - (7/14) \cdot .985 - (7/14) \cdot .592 \\ &= .151 \end{aligned}$$



$$\begin{aligned} \text{Gain}(S, \text{Wind}) &= .940 - (8/14) \cdot .811 - (6/14) \cdot 1.0 \\ &= .048 \end{aligned}$$

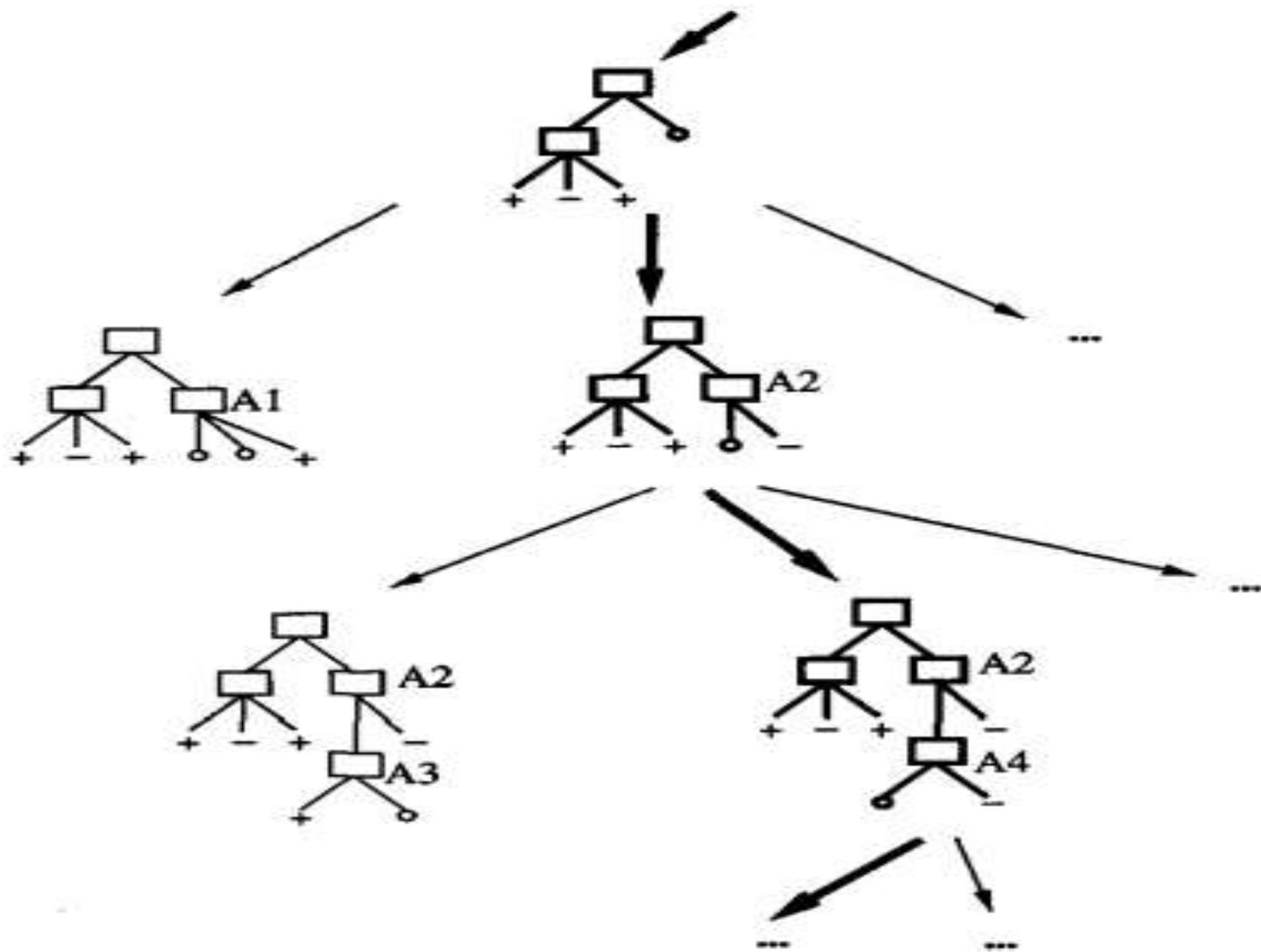


$$S_{\text{sunny}} = \{D1, D2, D8, D9, D11\}$$

$$\text{Gain}(S_{\text{sunny}}, \text{Humidity}) = .970 - (3/5) 0.0 - (2/5) 0.0 = .970$$

$$\text{Gain}(S_{\text{sunny}}, \text{Temperature}) = .970 - (2/5) 0.0 - (2/5) 1.0 - (1/5) 0.0 = .570$$

$$\text{Gain}(S_{\text{sunny}}, \text{Wind}) = .970 - (2/5) 1.0 - (3/5) .918 = .019$$

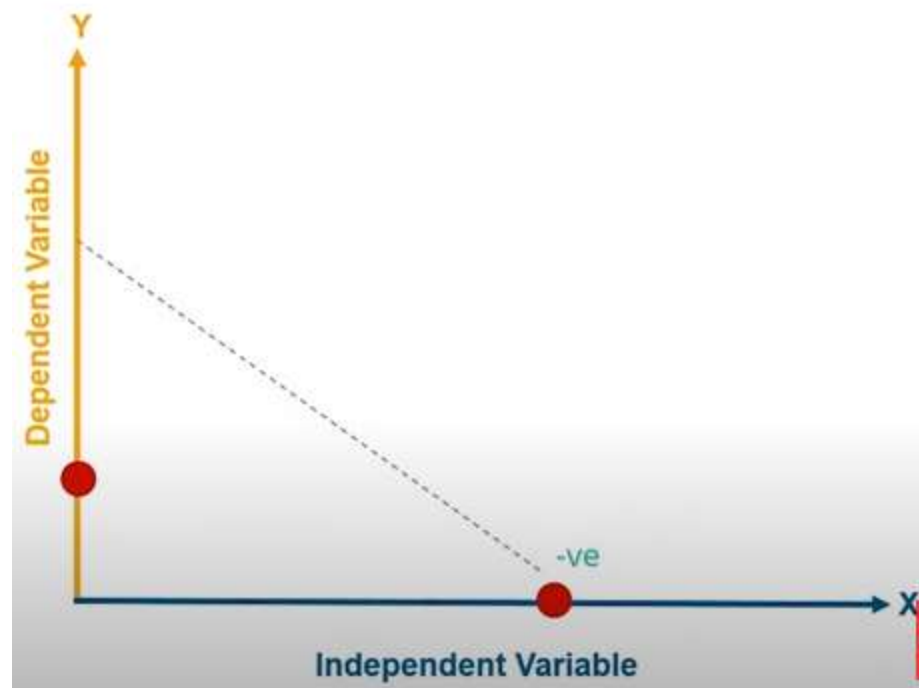
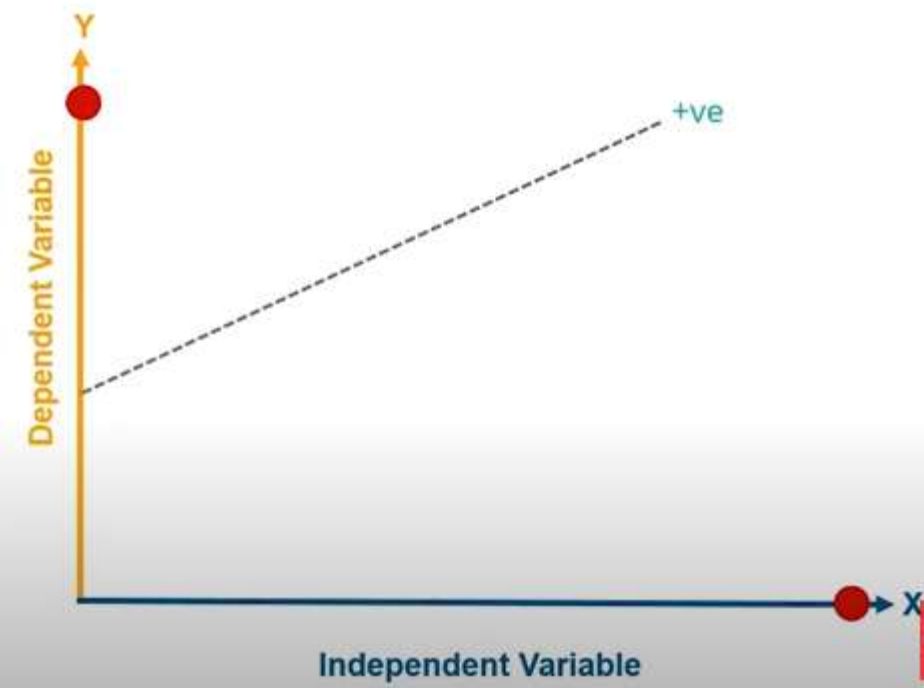


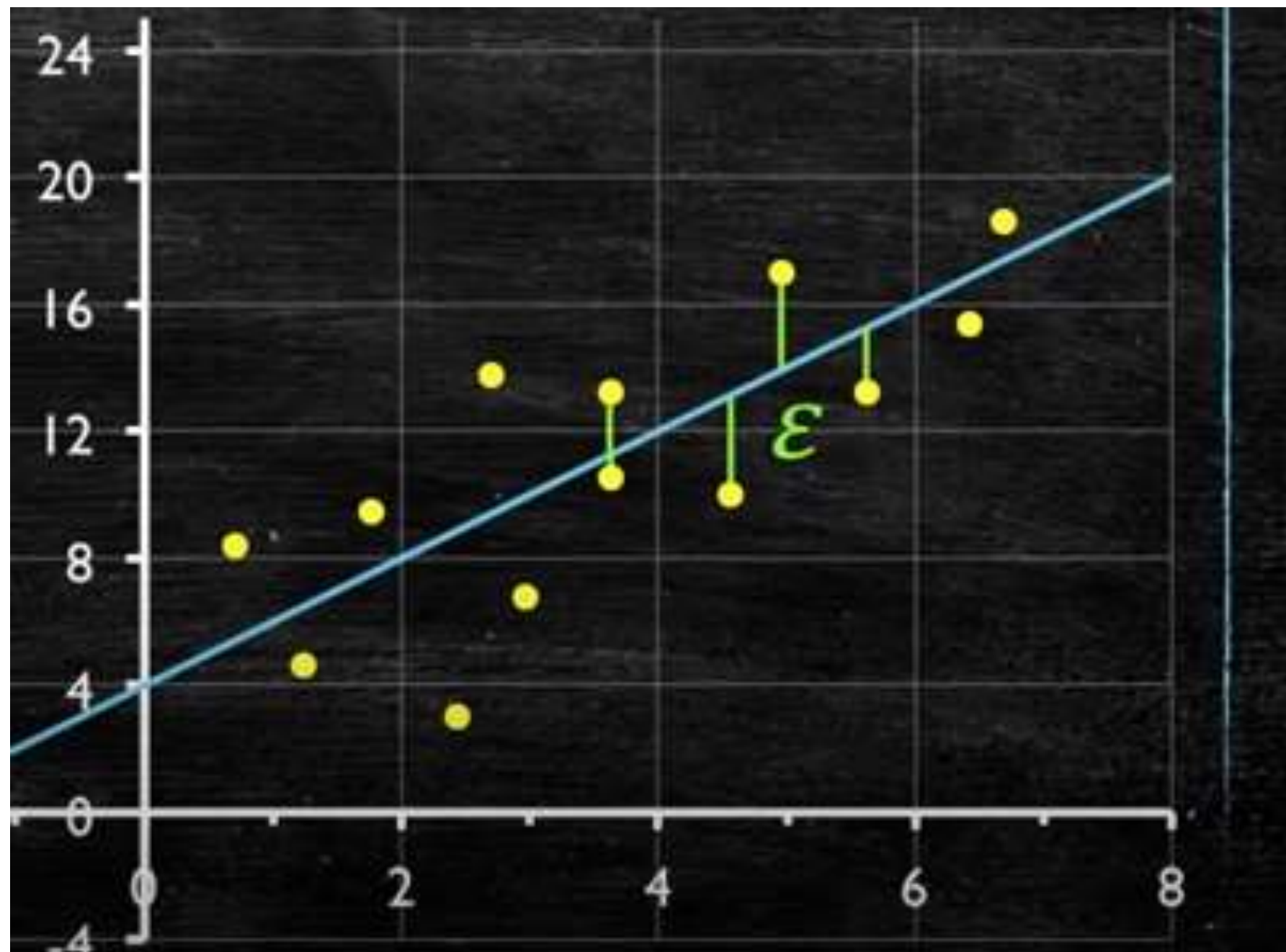
Linear Regression

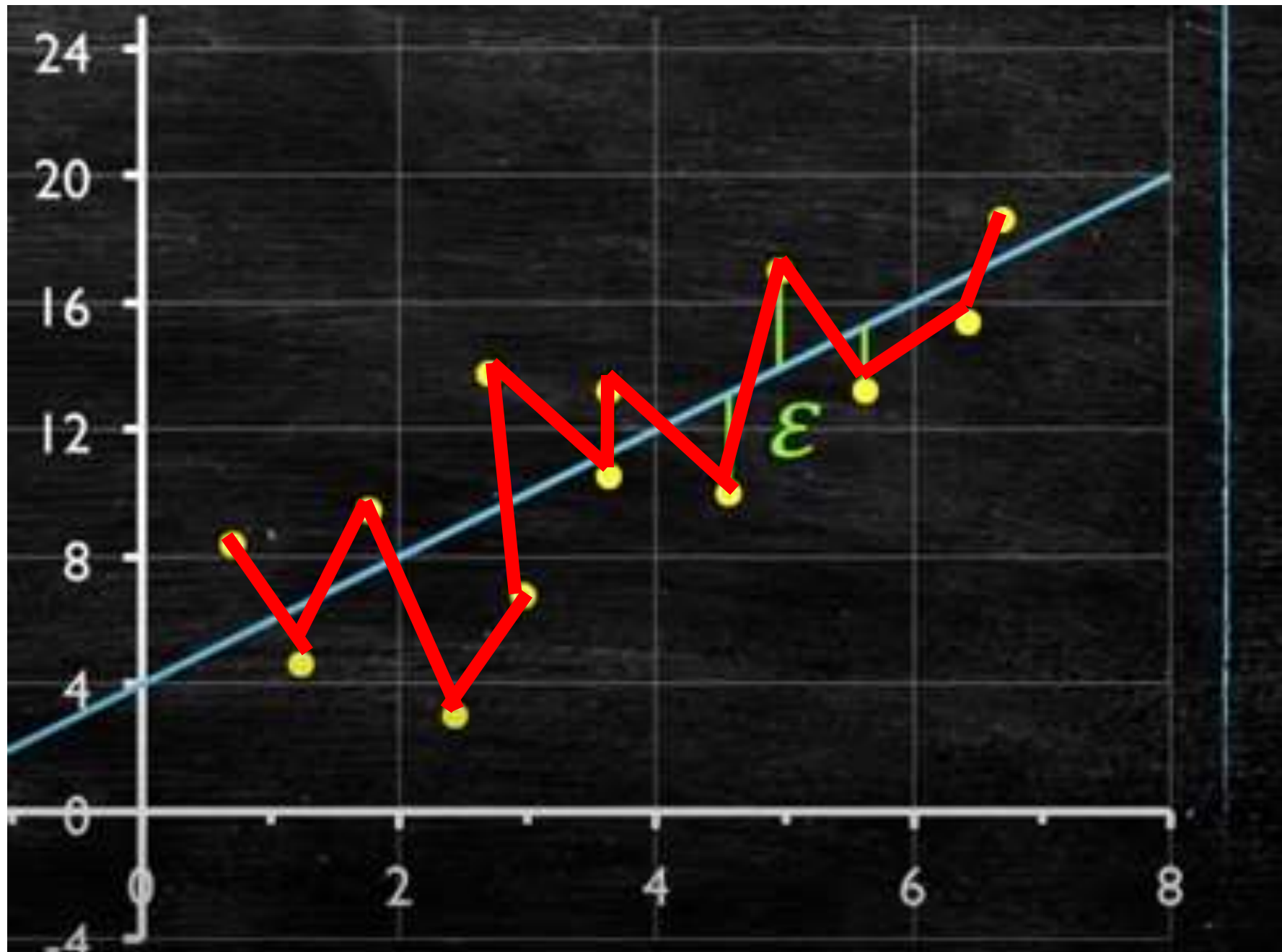
- Algorithm

- Analysis

- It is a statistical model used for predictive modeling in machine learning, in which an algorithm is used to predict continuous outcomes.
- Both classification & Regression capabilities.
- **Relates Dependent & Independent variables.**
- Forecast new observations...
 - (Ex: --Height vs weight
 - Income vs Expenses
 - Experience vs Salary
 - Shares & Stocks)

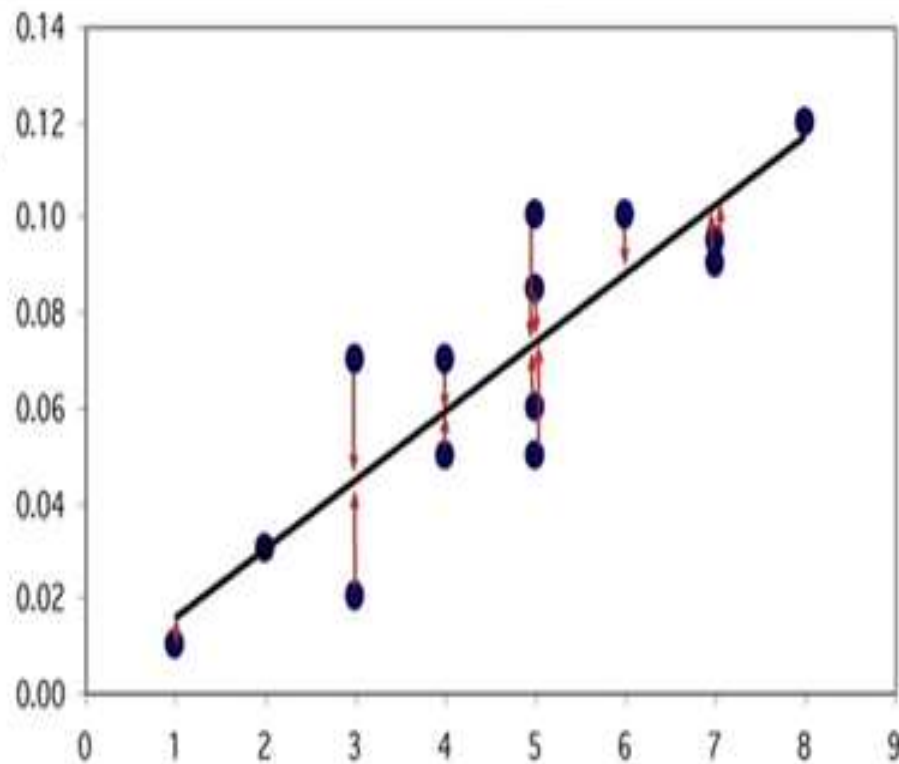






The regression line

The least-squares regression line is the unique line such that the sum of the squared vertical (y) distances between the data points and the line is the smallest possible.



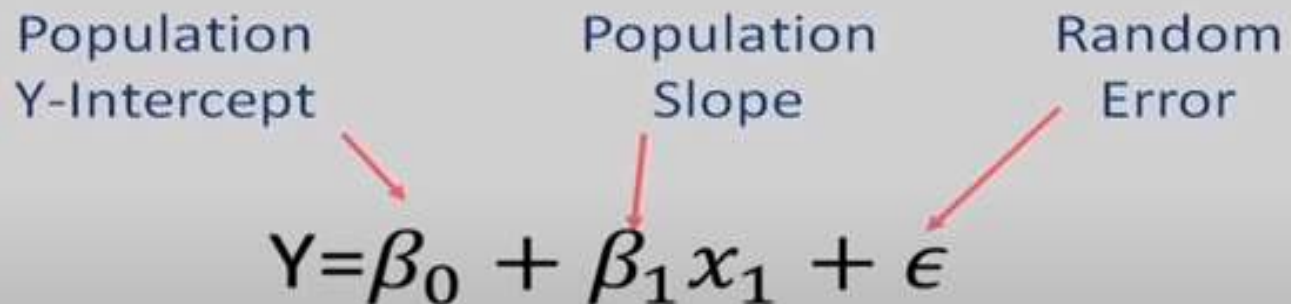
Linear Regression Model

- Relationship Between Variables Is a Linear Function

Population
Y-Intercept

Population
Slope

Random
Error

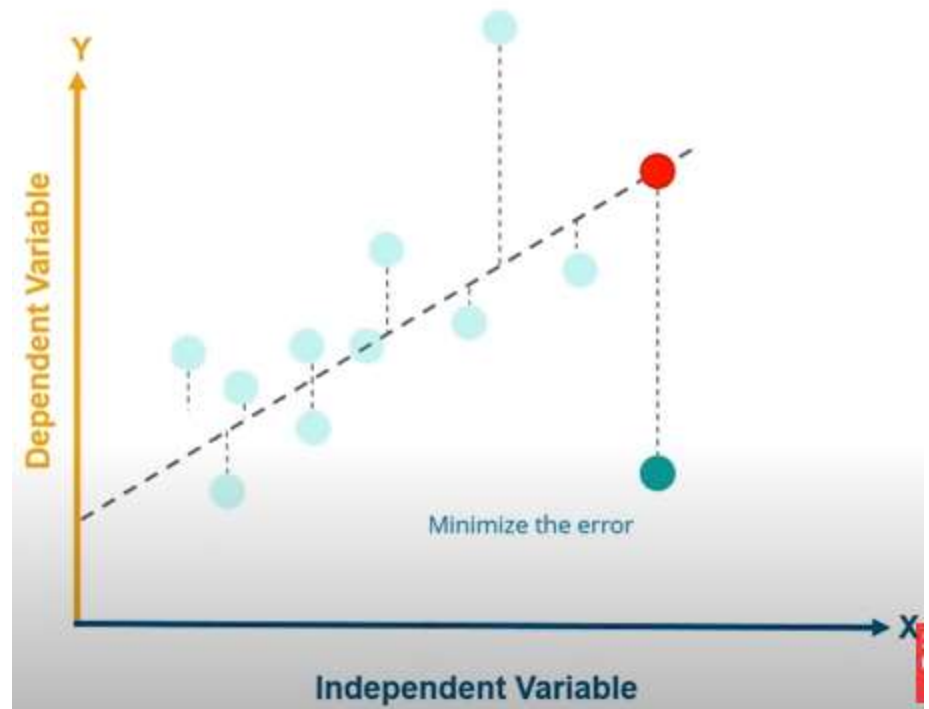
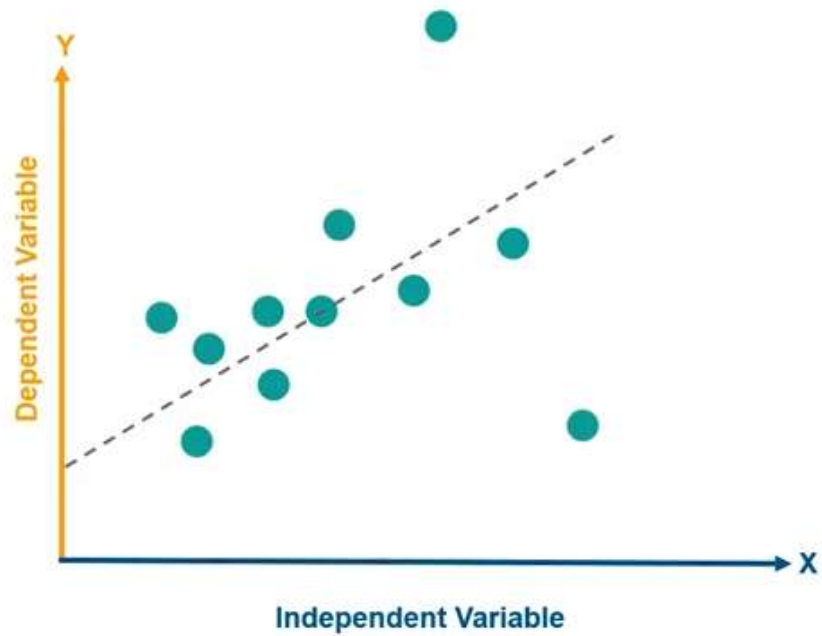


$Y = \beta_0 + \beta_1 x_1 + \epsilon$

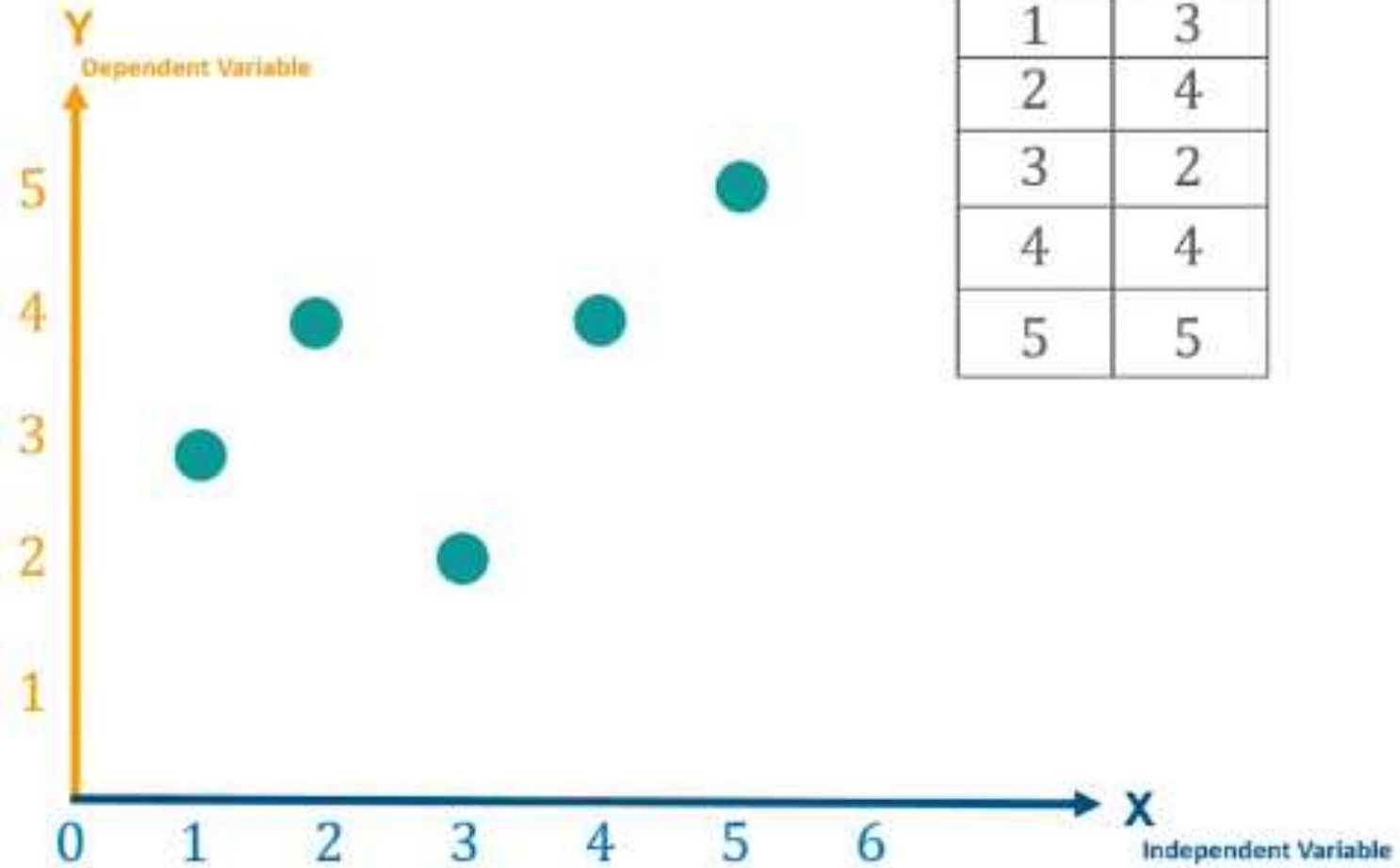
$$y = a + b x$$

$$y = m x + b$$

$$y = \beta_0 + \beta_1 x$$

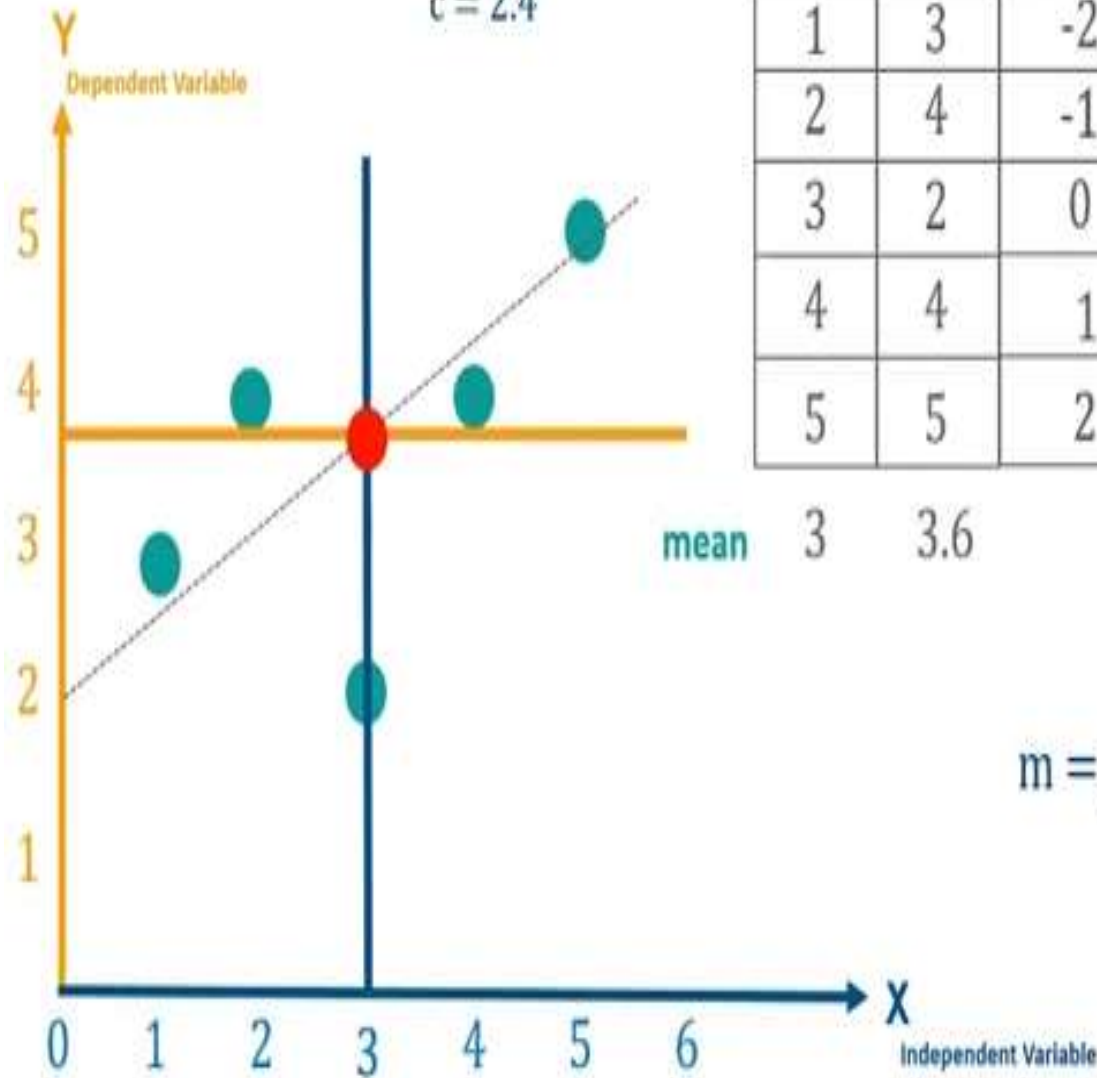


Example Data Set



$$y = mx + c$$

$$c = 2.4$$



x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
1	3	-2	-0.6	4	1.2
2	4	-1	0.4	1	-0.4
3	2	0	-1.6	0	0
4	4	1	0.4	1	0.4
5	5	2	1.4	4	2.8

$$\Sigma = 10 \quad \Sigma = 4$$

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{4}{10}$$

$$m = 0.4$$

$$c = 2.4$$

$$y = 0.4x + 2.4$$

$$m = 0.4$$

$$c = 2.4$$

$$y = 0.4x + 2.4$$

For given $m = 0.4$ & $c = 2.4$, lets
predict values for y for $x = \{1,2,3,4,5\}$

$$y = 0.4 \times 1 + 2.4 = 2.8$$

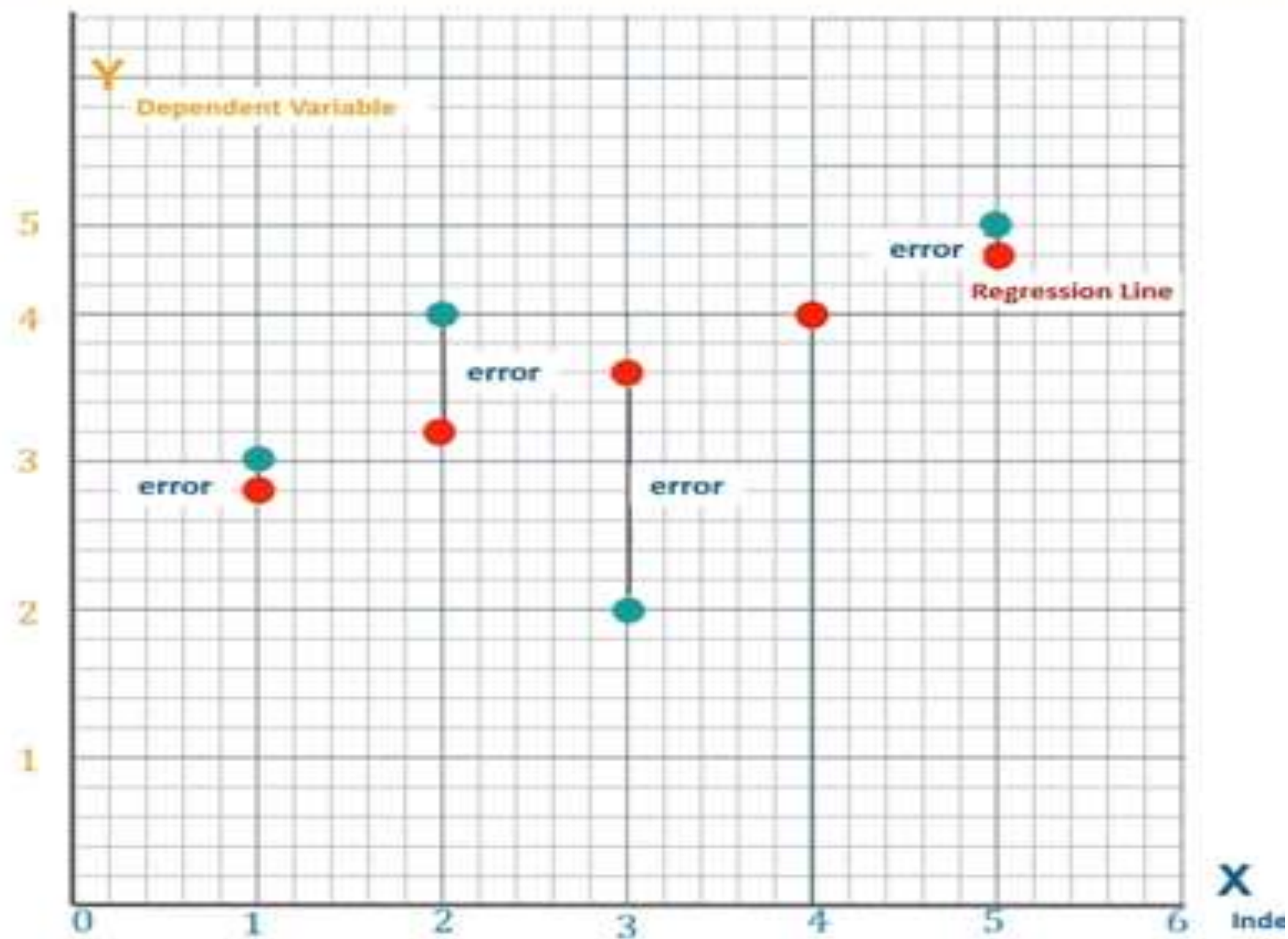
$$y = 0.4 \times 2 + 2.4 = 3.2$$

$$y = 0.4 \times 3 + 2.4 = 3.6$$

$$y = 0.4 \times 4 + 2.4 = 4.0$$

$$y = 0.4 \times 5 + 2.4 = 4.4$$

x	y
1	3
2	4
3	2
4	4
5	5

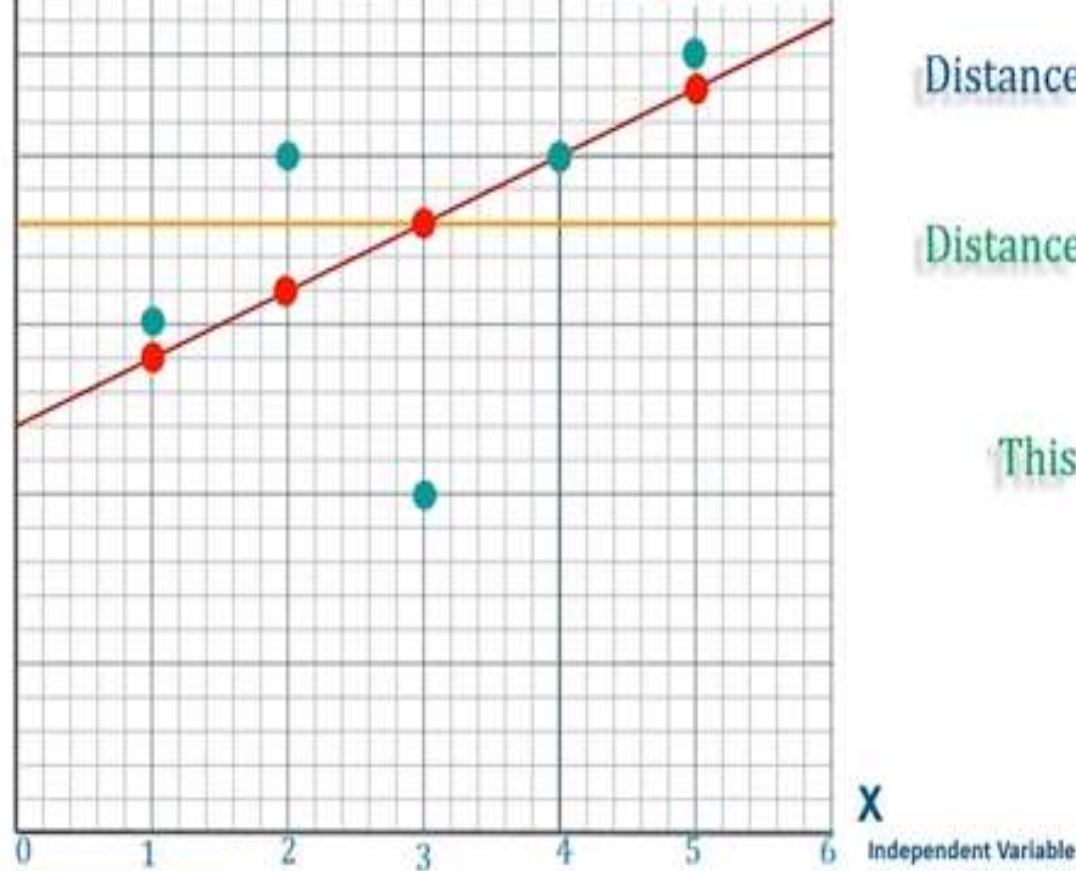


Actual vs Predicted Value

Y

Dependent Variable

Regression line



Distance actual - mean

vs

Distance predicted - mean

This is nothing but $R^2 = \frac{\sum (y_p - \bar{y})^2}{\sum (y - \bar{y})^2}$

Note : Error can be estimated with the above expression and it can also be reduced with the help of Gradient Descent Rule.

K-Nearest Neighbour

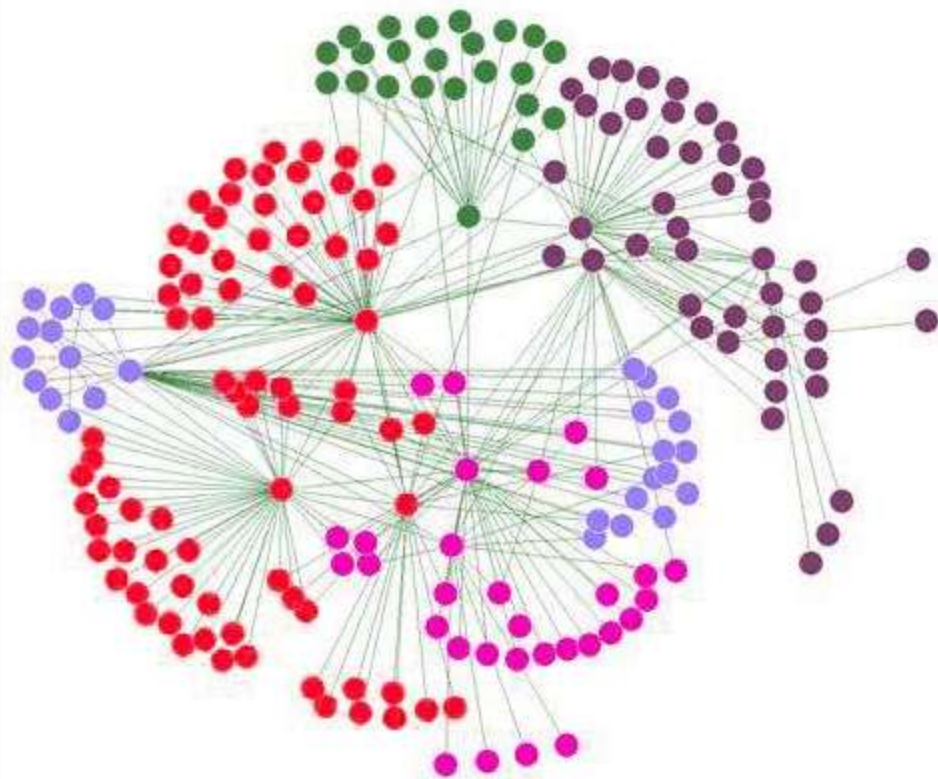
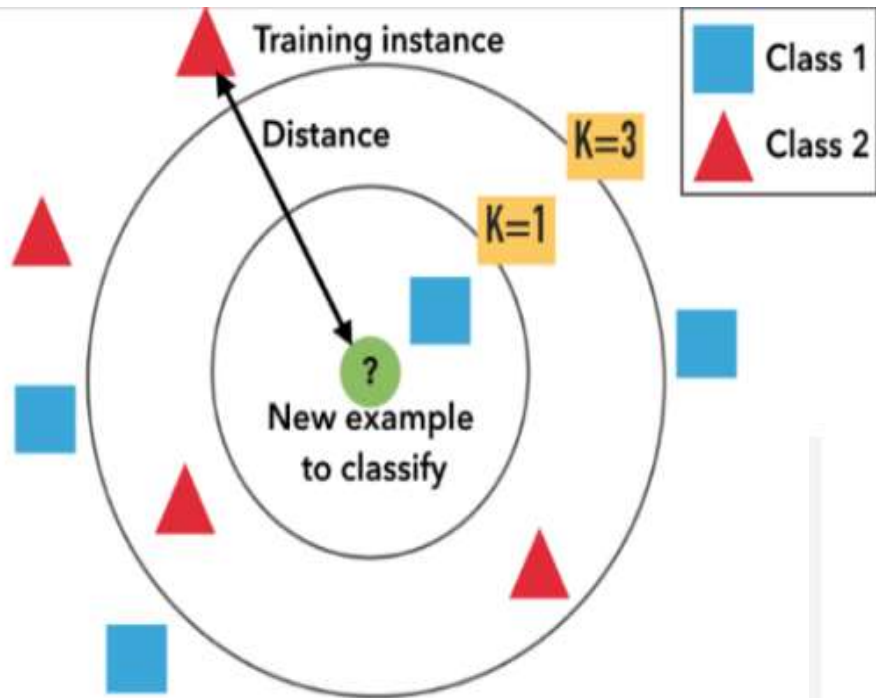
- The k-nearest neighbors (K-NN) algorithm is a simple, easy-to-implement supervised machine learning algorithm that can be used to solve both classification and regression problems.

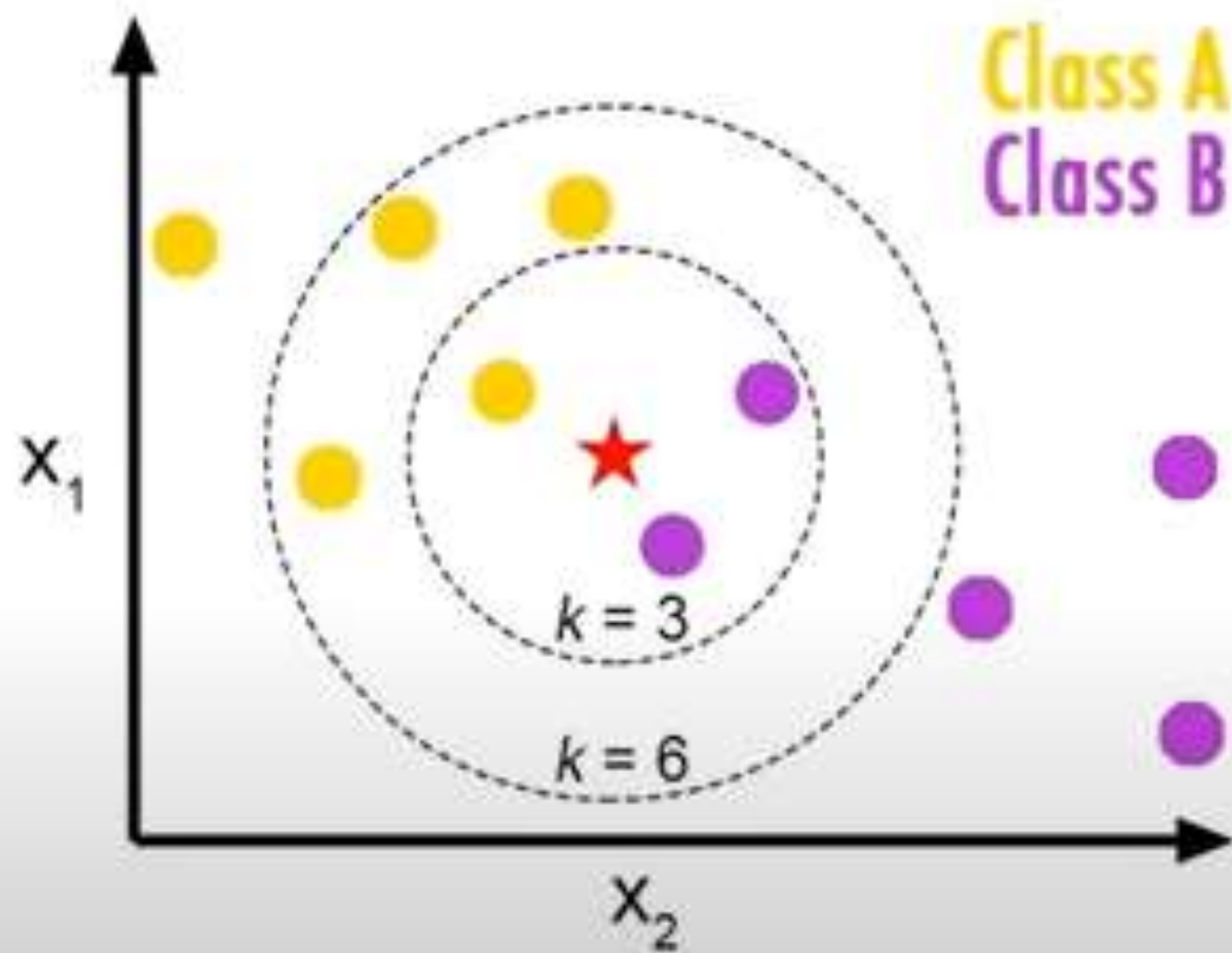
- K-NN is a non-parametric algorithm and stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm.

- It is also called instance based or lazy learner algorithm.

KNN Classifier







N	N	N
N	X	N
N	N	N

0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9

Basic k-nearest neighbor classification

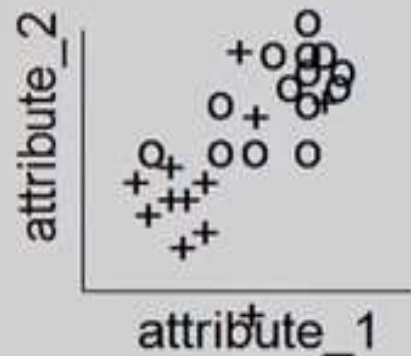
- Training method:
 - Save the training examples
- At prediction time:
 - Find the k training examples $(x_1, y_1), \dots, (x_k, y_k)$ that are closest to the test example x
 - **Classification**: Predict the most frequent class among those y_i 's.
 - **Regression**: Predict the average of among the y_i 's.

k-Nearest Neighbor

$$Dist(c_1, c_2) = \sqrt{\sum_{i=1}^N (attr_i(c_1) - attr_i(c_2))^2}$$

$$prediction_{test} = ?$$

- Average of k points more reliable when:
 - noise in attributes
 - noise in class labels
 - classes partially overlap



Algorithm :

- ❖ Choose the number K of the neighbors
- ❖ Calculate the Euclidean distance of K number of neighbors
- ❖ Based on Euclidean distance , take the K nearest neighbors as per the calculated.
- ❖ Among these k neighbors, count the number of the data points in each category.

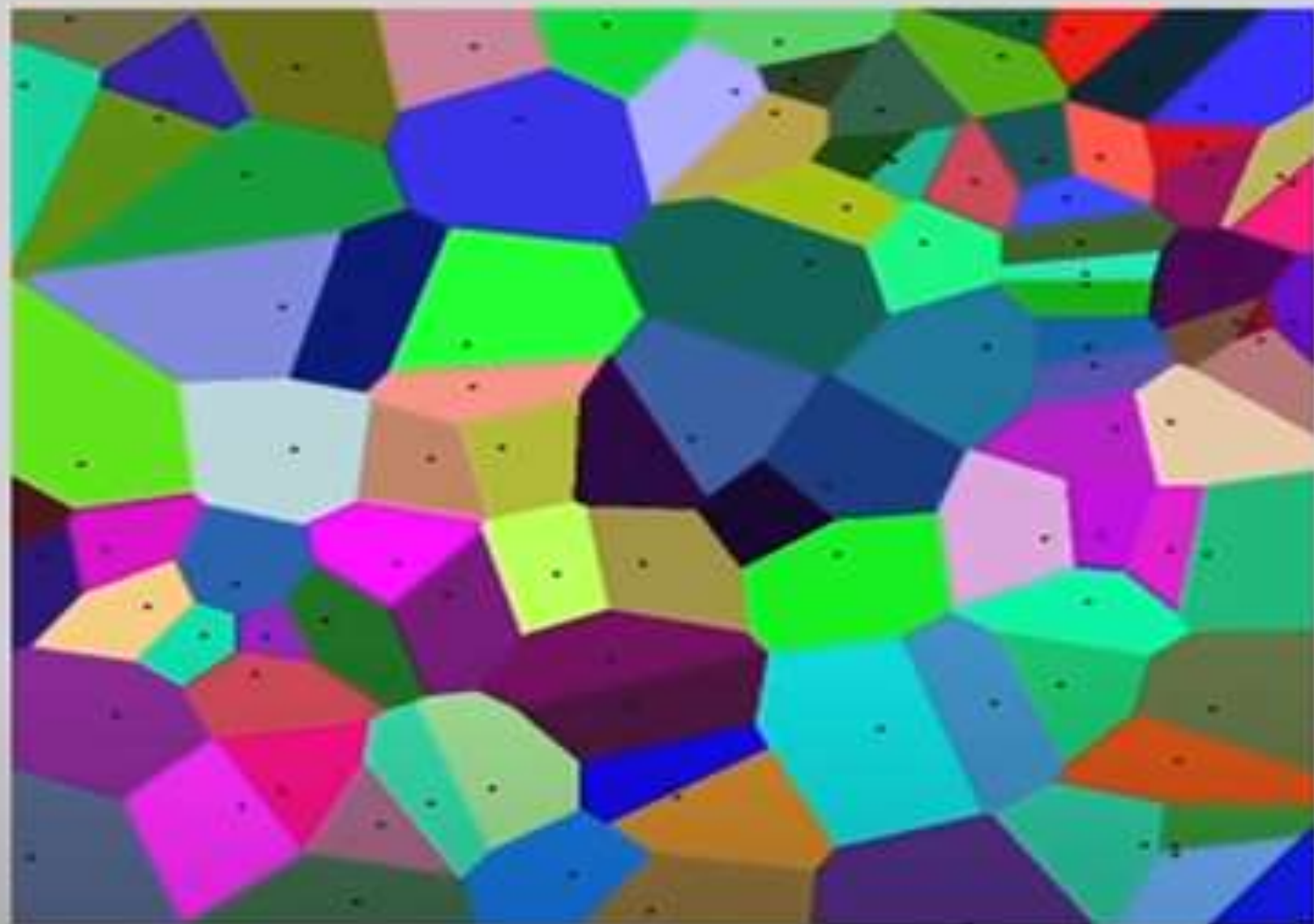
A	B	C	D	E	F	G	H
Height (in cms)	Weight (in kgs)	T Shirt Size	Euclidean Distance	Minimum Euclidean Distance			
158	58	M	4.242640687				
158	59	M	3.605551275				
158	63	M	3.605551275				
160	59	M	2.236067977	4			
160	60	M	1.414213562	1	New Data: Height: 161 cm		
163	60	M	2.236067977	3	Weight: 61 kg		
163	61	M	2	2			
160	64	L	3.16227766	5			
163	64	L	3.605551275				
165	61	L	4				
165	62	L	4.123105626		k = 5		
165	65	L	5.656854249				
168	62	L	7.071067812				
168	63	L	7.280109889				
168	66	L	8.602325267				
170	63	L	9.219544457				
170	64	L	9.486832981				
170	68	L	11.40175425				

Weighted Euclidean Distance

$$D(c1, c2) = \sqrt{\sum_{i=1}^N w_i \cdot \left(attr_i(c1) - attr_i(c2) \right)^2}$$

- large weights $\Rightarrow$ attribute is more important
 - small weights $\Rightarrow$ attribute is less important
 - zero weights $\Rightarrow$ attribute doesn't matter
-
- Weights allow kNN to be effective with axis-parallel elliptical classes
 - Where do weights come from?

Voronoi diagram



Age	W	Interested in game	Gender
10	35	Tennis	Male
15	40	Tennis	Female
20	52	Volleyball	Female
25	65	Volleyball	Male
30	70	Tennis	Male
35	75	Volleyball	Male
40	80	Tennis	Female
27	53	?	male

